

MARK YANG

(609)-580-0118 | yang.mar@northeastern.edu | Boston, MA | www.linkedin.com/in/markyang05 | https://mar-yan24.github.io/

EDUCATION

Northeastern University

Boston, MA (April 2028)

- BS in Electrical and Computer Engineering (3.98 GPA)
- Relevant Coursework: Embedded Design, Circuits and Signals, Fundamentals of Networks, Neuromechanical Simulation of Human Movement (MS level class), Differential Equations, Physics 2
- Activities: Putnam Club, Rock Climbing Club, Tau Beta Pi Engineering Honors Society

Queens University Belfast

Belfast, Northern Ireland (September–December 2024)

- N.U.*in* Study Abroad Semester

PROFESSIONAL EXPERIENCE

Philips Cambridge NA HQ

Software Test Engineer Co-op

Cambridge, MA (January 2026–Present)

- Validating the PIC iX event notification system with the Solution Reliability team, writing Gherkins to help define feature behavior and set up test cases for automation.
- Conducting robustness and reliability analysis of the alarm system for the Philips Telemetry Monitor through Azure DevOps workflow.
- Optimizing defect analysis workflows by engineering an internal LLM within the Azure cloud pipeline, helping automate feature vulnerability collation, reducing manual review time by 60-70% and providing real-time test coverage advice.

NEUMove

Undergraduate Researcher

Boston, MA (August 2025–Present)

- Programmed and implemented the compliant muscle tendon unit (CMTU) in the MuJoCo environment with PhD student Hyoungseo, and currently working on fitting the CMTU to existing OpenSim models to later integrate with the direct repository.
- Conducting MyoSuite simulations of normalized gait movement regardless of perturbation of the walking model to study variables causing major deviation in locomotion.

Northeastern University

Engineering Center Aide

Boston, MA (August 2025–January 2026)

- Worked as both hardware and software focus assistant for the freshman engineering class of 900+ students focusing on teaching and debugging tools such as Raspberry Pi Pico, Python, MATLAB, AutoCAD, and Solidworks.
- Maintained the first-year engineering makerspace 3D-printers and laser cutters giving consultations for optimized action, and hosted bi-weekly workshops for students on soldering.

Synbio Technologies

Software Engineering Intern

South Brunswick, NJ (June–August 2024)

- Improved user behavior visibility by developing a backend JavaScript event-logging system to track interactions, automated usage plots creation, allowing the team to identify and address UX bottlenecks.
- Conducted a complete overhaul of the main website, helping replace and modernize 100+ different pages of the main website using prior mentioned user behavior for focus points of improvement.

PROJECTS

Bipedal Spring Loaded Inverse Pendulum (bSLIP) Runner

Graduate Class Project (September 2025–Present)

- Implemented a two-leg bSLIP runner in MATLAB/Simulink (Simscape Multibody) with stance/swing PD control and contact/friction modeling.
- Performed apex-to-apex Poincare return-map analysis; added a one-step deadbeat regulator via touchdown hip-angle control to maintain apex height on rough terrain.

Hexapode Robot

Class Project (June–July 2025)

- Worked on controlling a Hexapode Robot using the DE1-SoC Intel FPGA board, designing a Quartus truth gate circuit for PWM servo control. Programmed a C++ script for the ARM System on a Chip for MMIO servo control, allowing the robot to walk, turn, dance, etc.

RoboCup Junior Maze Simulated Robot

Group Project (December 2021 – July 2022)

- Implemented a Breadth-First Search (BFS) navigation algorithm on the Erebus simulation platform and built custom maze worlds to systematically debug edge cases.
- Placed 1st in the 2022 RCJ international competition hosted in Thailand.

SKILLS AND HONORS

Languages/Tools Python, Java, C, C++, MATLAB, Simulink/Simscape, Git, Verilog, LTSpice, AWS

Awards/Honors 1st in Simulation Division 2022 RCJ Thailand, Morgan Stanley Early Insights